AMRITA VISHWA VIDYAPEETHAM | School of Engineering

Embedded Systems Project

Ghost Vehicle in Fog Detector Using Pressure Wave Sensing Before Headlights Appear

Mentor :

Dr. C. Ganesh Kumar

Chairperson, Department of Electronics and Communication Engineering, School of Computing, Chennai

Presented by :

Akshaya M V - CH.EN.U4CCE24003

Devadharshini P - CH.EN.U4CCE24004

M Mukesh - CH.EN.U4CCE24015



Ghost Sense Initiative

The Invisible Threat

- Dense fog critically reduces visual range on highways and rural roads, rendering standard headlights ineffective.
  Drivers face severely delayed detection of
- approaching vehicles, drastically increasing the likelihood of high-speed, catastrophic collisions.

The Edge-AI Solution

A New Paradigm

The Technology Stack

Zero-Latency Intelligence

SENSE

PROCESS

WARN

Detecting approaching vehicles before they become visible by capturing the low-frequency pressure waves they push ahead of them.

- Utilizes highly sensitive MEMS acoustic sensors. Processes physical data
- locally on a Cortex-M microcontroller.

- Deploys TinyML and Edge-AI directly on the chip. Triggers life-saving early warnings
- without relying on:



SENSE

Capture invisible pressure disturbances generated by moving vehicles.



PROCESS

Execute real-time signal analysis natively on a Cortex-M core.



WARN

Trigger immediate driver alerts through:

- Visual indicators (LEDs/Display)
- Audible alarms (Buzzer)

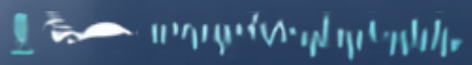
Architecture & Data Pipeline

Fromphysicalpressurewavestoreal-timeEdge-AI alerts.

Zone 1
Physical Sensing

Vehicle Approaches

Acoustic Detection



Signal Conditioning

Zone 2
Edge-AI Processing

4. Data Acquisition
5. Digital Filtering

6. Algorithmic Analysis

7. Threshold Decision

Zone 3
Alert Execution

8. Confirmation

9. Early Warning

10. Result

Pressure Waves > MEMS Sensor > Signal Conditioning > Cortex-M ADC > Digital Filtering > TinyML Analysis > Vehicle Detection > LED/LCD/Buzzer Alert > Driver Safety

Phase & Focus	Week	Core Activity	Key Milestone
Phase 1: Foundation(Research & Hardware)	Week 1-3	Literature Survey, Problem Analysis & Component Procurement.STM32/Cortex-M setup and architecture learning.	Hardware Procured & Environment Setup
Phase 2: Integration(Interfacing & Data)	Week 4 - 6	Sensor Interfacing & ADC Testing.Data Collection from live moving vehicles.Initial Signal Processing Implementation.	Successful Sensor Data Acquisition
Phase 3: Edge-AI(Algorithm & Alert)	Week 7 - 9	Vehicle Detection Algorithm (TinyML) Development.LED, Buzzer, and Display Integration.System Testing, Calibration, and Optimisation.	Functional Detection Algorithm
Phase 4: Finalisation(Prototype & Delivery)	Week 10 - 12	Complete Prototype Assembly.Result Analysis and Performance Evaluation.Report Writing and PPT Preparation.	Final Demonstration (Sept 20)

Hardware Ecosystem

Compute
- STM32F401 / STM32F103
- Cortex-M Microcontroller

Sensing
- MEMS Microphone Sensor
- MAX9814 / MAX4466

Conditioning
- LM358 Operational Amplifier
- Used for signal amplification

Output UI
- OLED Display / LCD (16×2)
- Warning LEDs
- Buzzer

Ancillary Components
- USB Programmer / ST-Link
- Breadboard
- Power Supply

Software Ecosystem

Environment
- STM32CubeIDE
- STM32CubeMX

Programming Language
- Embedded C Programming
- Optimized for TinyML constraints

Analysis Tools
- MATLAB
- Python
- Used for initial signal analysis and algorithm modeling

Simulation & Debugging
- Serial Terminal
  - PuTTY
  - Tera Term
- Proteus (optional for simulation)

Expected Output

Early Detection : Reliable identification of approaching vehicles completely obscured by dense fog.

Immediate Alert : Distinct visual and audible warnings delivered to the driver via the edge device with zero latency.

Ultimate Impact : Significantly improved road safety and a drastic reduction in collisions under critical low-visibility conditions.

Thank You !